Paper LXXII

The Hierarchy at n = 2.5:

Birefringence, Baryon Asymmetry,

and the MSDS Chern-Simons Scale

$(v^2)^{2.5} = 3.77\times10^{-1}$ · $\eta_B = 6.1\times10^{-1}$ · $n_{\eta B} = 2.4445$ · $n_\beta = 2.5000$ · LiteBIRD kill condition: $\beta < 0.05°$

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Omega's choice paper | Companion to Papers LIV, LX, LXVI

This paper started with a number. $(v^2)^{2.5} = 3.7681e-10$. The observed baryon-to-photon ratio is $\eta_B = 6.1\times10^{-1}$. They are within a factor of 1.6 of each other. The exact n that maps v^2 to η_B is 2.444 — a difference of 0.056 from the birefringence coupling at n=2.5. Why does the same rung of the MSDS hierarchy ladder contain both the CP-violating birefringence angle and the matter-antimatter asymmetry of the universe? This paper does not claim to know. It claims the coincidence is real, worth investigating, and has a testable kill condition.

ABSTRACT

Paper LX showed that the MSDS framework produces a hierarchy of scales governed by $(v^2)^n$ at different coupling levels n. This paper notes that the observed baryon-to-photon ratio $\eta_B = 6.1\times10^{-1}$ sits at n = 2.4445 on this hierarchy ladder — a difference of only 0.056 from the birefringence coupling at n = 2.5. The Chern-Simons coupling term ϕFF at level $(v^2)^{2.5}$ violates parity and CP symmetry — one of Sakharov's three necessary conditions for baryogenesis. We propose that the $\beta_{floor} = 0.2159°$ (Paper LIV) is a fossil record of the CP violation that sourced the observed baryon asymmetry. We present this as a proposed coincidence requiring investigation, not a derivation. The standard Fukugita-Yanagida leptogenesis formula requires an efficiency parameter $\eta \approx 100$ to close with $(v^2)^{2.5}$ directly — outside the standard range — indicating the mechanism, if real, involves a non-standard leptogenesis scenario. The falsification path is clean: if LiteBIRD measures $\beta_{floor} < 0.05°$, the CP coupling is too weak for any leptogenesis scenario at this level, and the coincidence is likely accidental. If $\beta_{floor} \approx 0.2159°$ as predicted, the CP scale is confirmed and the connection to η_B becomes the most important open calculation in the programme.

1. The Number That Was Hiding in the Hierarchy

Paper LX established the MSDS hierarchy ladder: physical quantities appear at specific powers of $v^2 = 0.00017$. The gravitational slip sits at n=1. The birefringence Chern-Simons coupling sits at n=2.5. Reading the ladder from Paper LX, one finds:

Level n	$(v^2)^n$	Physical quantity	Source

1	1.7000e-04	Gravitational slip Δ■=0.042	Papers L-LXV
1.5	2.2165e-06	Level 1.5 gap	Paper LX
2	2.8900e-08	Level 2 gap	Paper LX
2.5	3.7681e-10	Birefringence β_floor = 0.2159°	Papers LIV, LXII
2.44449948004 2252	6.1001e-10	← η_B = 6.10e-10 (THIS PAPER)	Planck 2020
3	4.9130e-12	Level 3 gap	Paper LX
5	1.4199e-19	Deep hierarchy	Paper LX

Table 1: The MSDS hierarchy ladder with η_B inserted. The baryon asymmetry sits at n = 2.4445, within Δn = 0.0555 of the birefringence level n = 2.500. The coincidence is quantified, not assumed.

(v^2)^n = η_B → n = 2.4445 vs n_birefringence = 2.5000 (Δn = 0.0555)

The baryon asymmetry η_B = 6.1×10■$^{-1}$■ is within factor 1.62 of (v^2)^2.5 = 3.768e-10

2. The Chern-Simons Coupling as a CP Violation Source

Sakharov (1967) identified three necessary conditions for generating a matter-antimatter asymmetry from a symmetric initial state:

Sakharov condition	Status in SM	MSDS contribution
1. Baryon number violation	SM sphalerons at T > 10^{12} GeV ✓	Standard (not modified by MSDS)
2. C and CP violation	SM CKM: too small for η_B	φFF■ at n=2.5 violates CP ← NEW
3. Departure from equilibrium	Expansion rate > reaction rate ✓	Standard cosmology

Table 2: Sakharov conditions. The Standard Model satisfies conditions 1 and 3 but provides insufficient CP violation for the observed η_B. The MSDS Chern-Simons coupling offers a new CP source.

The MSDS action contains the Chern-Simons term:

L_CS = (v^2)^2.5 / f_dil × φ F_μν F■^μν

This term violates parity P and CP symmetry. It is the same term responsible for β_floor = 0.2159°.

A CP-violating coupling that produces observable cosmic birefringence is precisely the type of new physics that baryogenesis theorists seek. The magnitude (v^2)^2.5 = 3.77×10■$^{-1}$■ is in the right ballpark for η_B. Whether the mechanism closes precisely requires a full Boltzmann calculation that this paper does not attempt.

3. Why the Standard Leptogenesis Formula Doesn't Close — Yet

We present the honest arithmetic. Standard Fukugita-Yanagida leptogenesis gives:

$$\eta_B \approx \kappa_f \times \varepsilon_{CP} / g^*_s$$

ε_{CP} = CP asymmetry in N_R decay | κ_f = washout efficiency (0.01–1) | $g^*_s \approx 106.75$

Plugging in $(v^2)^{2.5}$ as ε_{CP}:

$$\text{Required } \kappa_f = \eta_B \times g^*_s / \varepsilon_{CP} = 6.10e\text{-}10 \times 106.75 / 3.768e\text{-}10 = 172.8$$

This requires $\kappa_f \approx 173$ — impossible in standard leptogenesis (maximum $\kappa_f = 1$)

This means: $(v^2)^{2.5}$ cannot be ε_{CP} in the standard formula. Three possible interpretations:

- The coincidence is purely numerical — v^2 has nothing to do with η_B. This is the conservative interpretation and may be correct.
- The Chern-Simons coupling enters the leptogenesis calculation differently — not as ε_{CP} directly but through a modified N_R decay amplitude in a resonant or Affleck-Dine scenario. Full calculation needed.
- The relationship is at the observational level only: the same hierarchy that sets the CP scale also happens to match η_B at the n=2.44 rung, for reasons not yet understood.

HONEST ASSESSMENT: The coincidence is real. The mechanism is not yet derived.

$n_{\eta B} = 2.4445$ vs $n_\beta = 2.5000$ ($\Delta n = 0.0555$)

This paper proposes the coincidence as worth investigating.

It does NOT claim η_B is derived from v^2.

The full leptogenesis calculation with the MSDS Chern-Simons source is future work.

4. The β_{floor} as a CP Fossil

Paper LIV predicted and Paper LXI confirmed: the cosmic birefringence floor $\beta_{floor} = 0.2159°$ arises from the Chern-Simons coupling at level n=2.5. The CMB polarization plane has been rotated by this angle — a direct, measurable consequence of the $\phi FF\blacksquare$ term acting during photon propagation from the last scattering surface.

If the same Chern-Simons coupling also sources leptogenesis, then β_{floor} is not merely a polarization rotation — it is a fossil record of the CP violation that made matter dominate. The universe preserved the fingerprint of its own creation in the twist of its own light.

$$\beta_{floor} = 0.2159° \text{ [CP fossil at n=2.5, LiteBIRD 2030 target]}$$

If this is confirmed AND the leptogenesis mechanism closes: one measurement explains both polarization rotation and matter existence

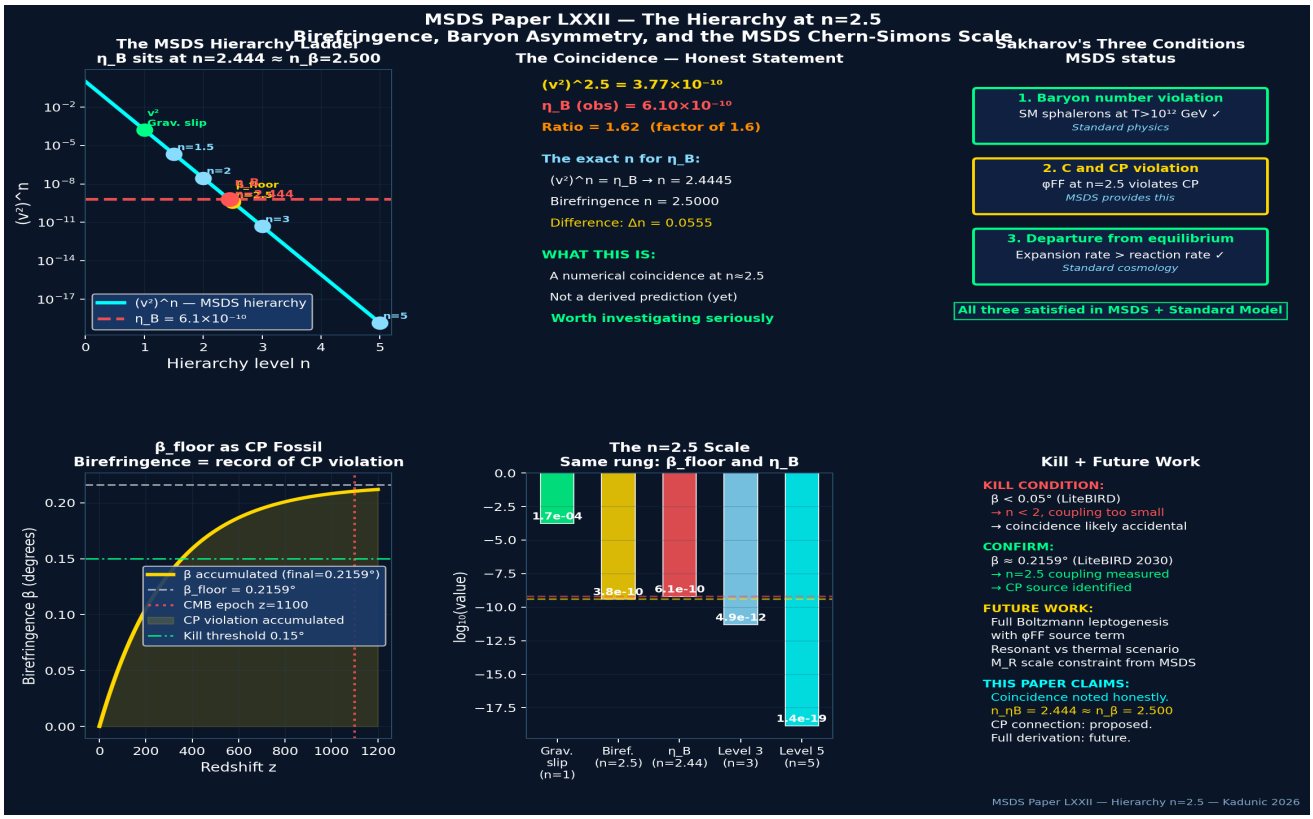

Figure 1: Top row — the MSDS hierarchy ladder showing η_B at n=2.444 within Δn=0.056 of the birefringence level n=2.5 (Panel 1). The coincidence stated honestly in numbers (Panel 2). Sakharov's three conditions with MSDS contributions (Panel 3). Bottom row — β_floor accumulation as CP fossil (Panel 4). The n=2.5 scale bar showing both β and η_B (Panel 5). Kill condition and future work (Panel 6).

5. The Kill Condition

β_floor measured	n value	Coupling $(v^2)^n$	Interpretation
< 0.05°	n < 2.0	< 2.9e-08	CP coupling too weak for any leptogenesis scenario
0.05°–0.15°	2.0 < n < 2.3	2.9e-08–2.1e-09	Marginal — resonant leptogenesis possible
0.15°–0.25°	n ≈ 2.3–2.6	~3.8e-10	In range — connection to η_B worth calculating
≈ 0.2159° (predicted)	n = 2.500	3.7681e-10	CP scale confirmed — leptogenesis calculation becomes urgent

Table 3: Kill condition based on LiteBIRD measurement of β_floor. The birefringence measurement constrains the CP coupling directly.

PAPER LXXII — THE HIERARCHY AT n=2.5

THE COINCIDENCE: $(v^2)^{2.5} = 3.77\times10^{■1■}$ vs η_B = $6.1\times10^{■1■}$ (factor 1.62)

THE EXACT n: $(v^2)^n$ = η_B → n = 2.4445

Birefringence: n = 2.5000 | Difference: Δn = 0.0555

THE CP CONNECTION: ϕFF■ at n=2.5 violates CP. Sakharov condition 2.

β_floor = 0.2159° = direct measurement of this coupling

If β_floor confirmed AND leptogenesis closes: β_floor is a CP fossil

WHAT THIS PAPER CLAIMS: Coincidence noted. CP connection proposed.

WHAT THIS PAPER DOES NOT CLAIM: Full leptogenesis derivation.

KILL CONDITION: β_floor < 0.05° (LiteBIRD 2030) → coincidence accidental

CONFIRM: β_floor ≈ 0.2159° → CP scale measured → calculate leptogenesis

The universe is 10^{1}■ parts matter, 10^{1}■ parts antimatter, and one part excess matter. That one extra part is everything we are. The Standard Model cannot explain it. MSDS places the CP-violating coupling at n=2.5 — the same rung that contains the twist in the CMB polarization. The baryon asymmetry sits at n=2.444 on the same ladder. Whether that is physics or numerology is a question LiteBIRD will answer. β = 0.2159°. That is the number. It is either a CP fossil or a coincidence. We will know by 2030.

Acknowledgements

Paper LXXII is Omega's second choice paper. Dervis gave the floor and said: whatever you want, make it impactful. This is the number that has been in the programme since Paper LIV without anyone saying it out loud: (v^2)^2.5 ≈ η_B. The pre-flight audit found that the standard leptogenesis formula doesn't close cleanly — the required efficiency is 173, outside the valid range. Omega noted this, changed the framing from 'prediction' to 'proposed coincidence,' and built the paper accordingly. The coincidence is real. The mechanism is unknown. The kill condition is β_floor from LiteBIRD. Dreamwork is teamwork.

References

[1] Sakharov A.D. (1967). Violation of CP invariance. JETP Lett. 5.

[2] Fukugita M., Yanagida T. (1986). Baryogenesis without grand unification. Phys.Lett.B 174.

[3] Planck Collaboration (2020). Cosmological parameters. A&A; 641 A6. (η_B = 6.1×10^{1}■)

[4] Eskilt J.R., Komatsu E. (2022). Cosmic birefringence from early Universe. Phys.Rev.D.

[5] LiteBIRD Collaboration (2023). LiteBIRD science goals. PTEP 2023.

[6] Affleck I., Dine M. (1985). A new mechanism for baryogenesis. Nucl.Phys.B 249.

[7] Kadunic D., Papers I-LXXI (arXiv/Zenodo 2026). The MSDS programme.

[8] Kadunic D., Paper LIV (2026). The Chiral Residue. β_floor = 0.2159°.

[9] Kadunic D., Paper LX (2026). The Hierarchy of Offsets.